ECEN 248 - Lab Report

Lab Number: 3

Lab Title: Rudimentary Adder Circuits

Section Number: 509

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Date: 9/28/2022

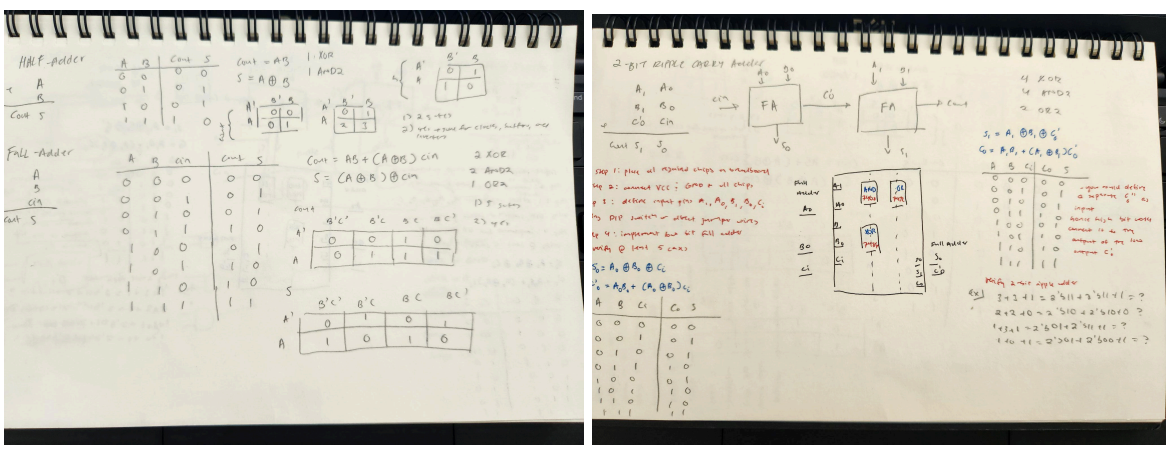
TA: Sri Kodi

Objectives:

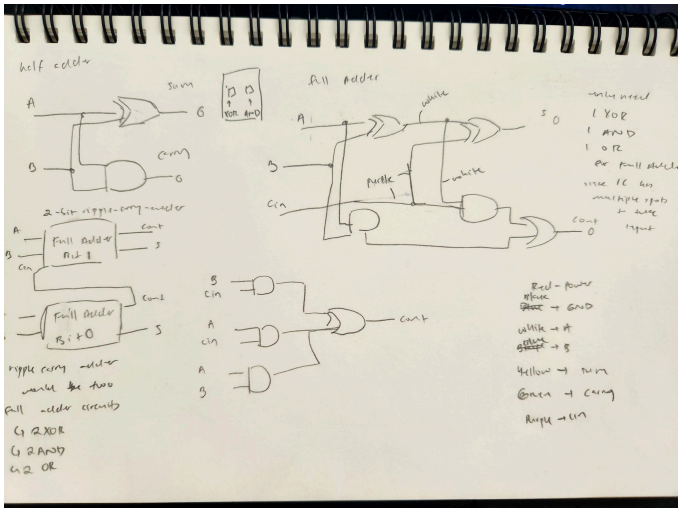
The purpose of this lab is to design simple combination adder circuits including a half adder, full adder, and ripple carry adder. By doing this, the idea of Karnaugh-Maps is reinforced while an understanding of simple arithmetic logic is gained. The lab will demonstrate how to build adder circuits using breadboards and how to test/debug them till the circuit works following the corresponding truth table.

Design:

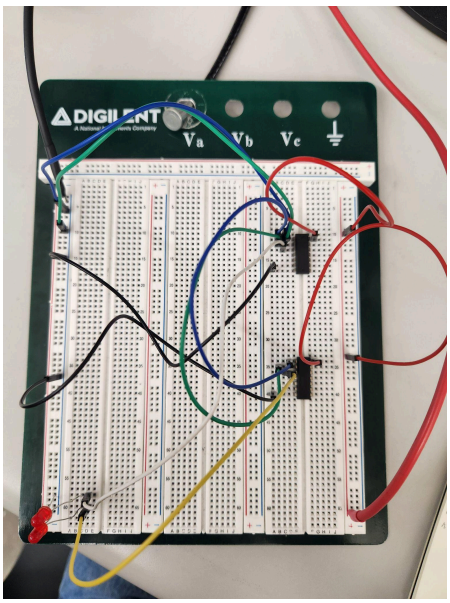
In order to complete this lab, circuit diagrams and truth tables were made for the half adder, full adder, and ripple carry adder incorporating the minimum appropriate ICs. From there, each circuit was built using a breadboard, tested, and the output was compared to the truth table. For the half adder, an XOR and AND gate were used. For the full adder, an XOR, AND, and OR gate were used. Then for the ripple carry adder, two full adders were combined resulting in the use of 2 XOR, 2 AND, and 2 OR gates.

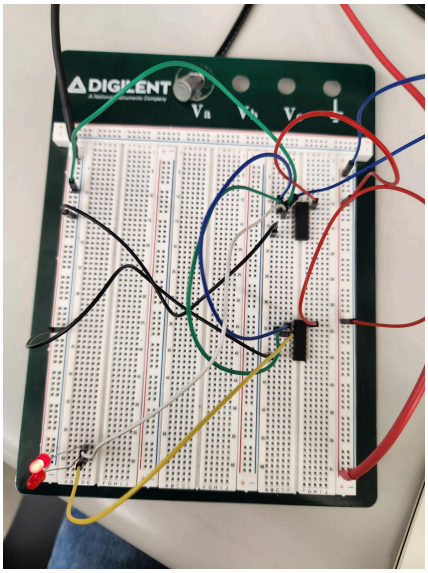


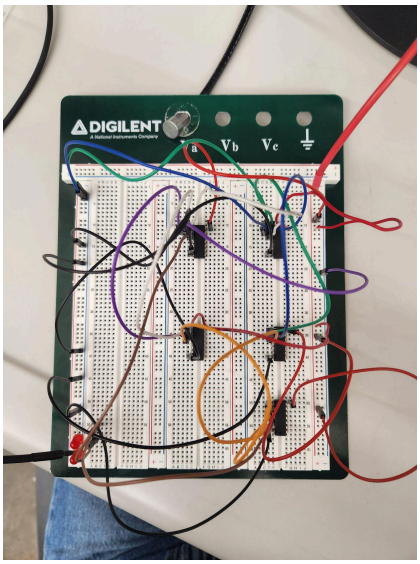
Designs and truth tables for carry ripple adder, half adder, and full adder



Circuit Designs for half adder and full adder





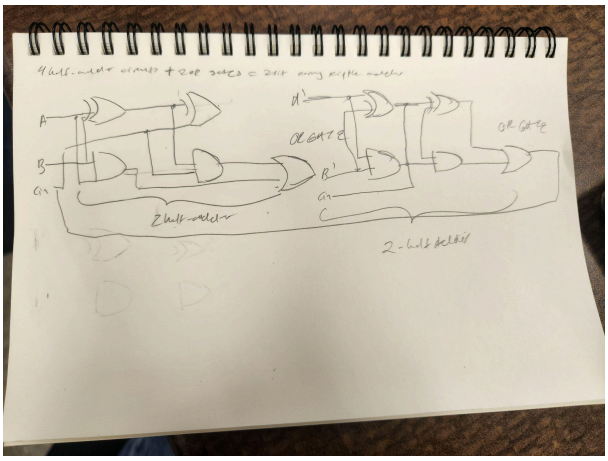
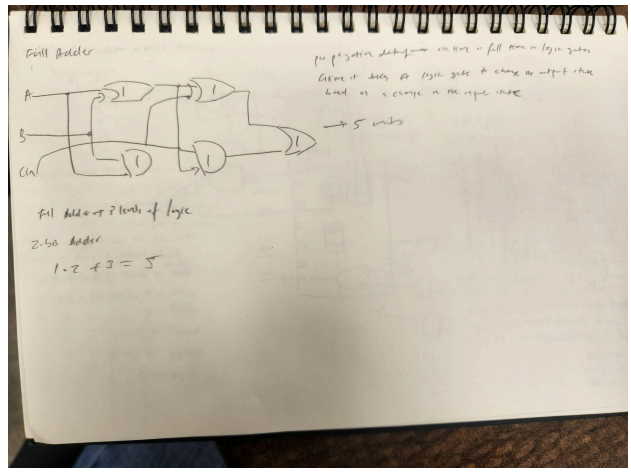


Breadboard design for half adder and full adder

Results:

The designs for each of the adder circuits were executed correctly, though both the full adder and carry ripple adder's designs were modified to be simpler. At first, the full adder was using two XOR gates, two AND gates, and one OR gate, but was then simplified to be one gate for each. Since the IC is able to take several inputs and outputs, there was no need for the extra XOR and AND gate. The carry ripple adder was then simplified as well, due to it being based on the design for the full adder.

Conclusion:

Through this lab, I was able to gain an understanding of how full, half, and ripple carry adders work by designing, implementing, and testing the circuits using a breadboard. I learned how to simplify circuits using logic gates and how to use two full adder circuits to make a ripple carry adder.

Post-lab Deliverables: